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SDR DDS Solution Design

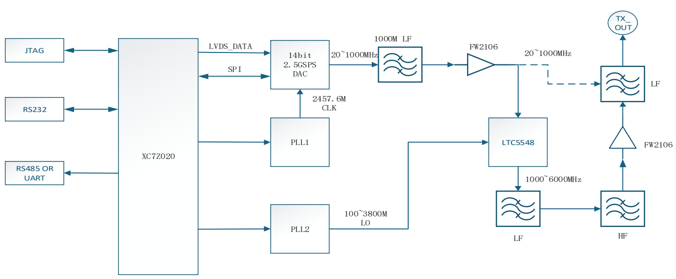
Overview

The digital baseband of the SDR DDS adopts an intermediate frequency architecture combining FPGA and DAC to generate digital signals. The FPGA model is XC7Z020, and the DAC features a sampling rate of 2.5 GSPS with a 14-bit resolution. The digital subsystem is capable of generating baseband signals ranging from 20 MHz to 1000 MHz with a maximum signal bandwidth of 500 MHz. Equipped with an internal PLL and mixer, this module upconverts the baseband signals to the frequency band of 1000 MHz to 6000 MHz.

Product Features

Supports arbitrary IQ-modulated wideband intermediate frequency signals (OFDM / chirp / pulse)。
500 MHz ultra-wide bandwidth, multi-carrier capability, Real-time FPGA update with nanosecond-level latency, Software-configurable IF frequency, signal bandwidth and filtering modes。

Design Scheme



The XC7Z020 generates baseband IQ signals, and implements FIR interpolation plus CIC upsampling to meet the IF sampling rate requirement of the DAC. Pulse-shaping filters are deployed to suppress out-of-band spurs. Afterwards, the baseband IQ signals are modulated onto the configured digital IF carrier to complete frequency up-conversion and generate real digital IF signals. The I and Q components are combined into a single real IF waveform and sent to the DAC. Integrated with a dual-core Cortex-A9 processor, the XC7Z020 receives commands from the host computer and controls the operating modes of the DAC and all PLLs.

The DAC integrates built-in sinc roll-off filtering, and is matched with external LC or ceramic IF band-pass filters to eliminate DAC sampling images and high-frequency harmonic spurs. It converts the real digital IF sampling codes output by the FPGA into continuous analog IF voltage waveforms. This DAC operates at a sampling rate of 2.5 GSPS with 14-bit resolution, and can generate analog IF waveforms ranging from 20 MHz to 1000 MHz.

The DAC requires a 2457.6 MHz input clock, which is supplied by a PLL.

The LTC5548 is a microwave mixer with a wideband IF coverage from DC to 6 GHz and RF coverage of 2 GHz to 14 GHz. Its LO input features an integrated ×2 frequency multiplier. This allows the PLL2 output frequency range to be reduced, enabling RF signal output ranging from 1000 MHz to 6000 MHz.

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